

Environment Parameters

Exported from: env.yaml

[Scene Robot Actuators]

<i>actuator</i>	<i>property</i>	<i>value</i>
legs	class_type	isaacsim.actuators.actuator_pd:ImplicitActuator
	joint_names_expr	['FL_hip_joint', 'FR_hip_joint', 'FL_thigh_joint', 'FR_thigh_joint', 'FL_calf_joint', 'FR_calf_joint']
	effort_limit	60.0
	velocity_limit	20.0
	effort_limit_sim	None
	velocity_limit_sim	None
	stiffness	40.0
	damping	2.0
	armature	None
	friction	None
	dynamic_friction	None
	viscous_friction	None
wheels	class_type	isaacsim.actuators.actuator_pd:ImplicitActuator
	joint_names_expr	['FL_foot_joint', 'FR_foot_joint']
	effort_limit	12.0
	velocity_limit	12.5
	effort_limit_sim	None
	velocity_limit_sim	None
	stiffness	0.0
	damping	1.0
	armature	None
	friction	None
	dynamic_friction	None
	viscous_friction	None

[Actions]

<i>action</i>	<i>property</i>	<i>value</i>
hip_joint_pos	class_type	isaacclab.envs.mdp.actions.joint_actions:JointPositionAction
	asset_name	robot
	debug_vis	False
	clip	None
	joint_names	FL_hip_joint
		FR_hip_joint
	scale	1.0
	offset	0.0
	preserve_order	True
	use_default_offset	True
shoudler_thigh_joint_pos	class_type	isaacclab.envs.mdp.actions.joint_actions:JointPositionAction
	asset_name	robot
	debug_vis	False
	clip	None
	joint_names	FL_thigh_joint
		FR_thigh_joint
		FL_calf_joint
		FR_calf_joint
	scale	1.0
	offset	0.0
	preserve_order	True
	use_default_offset	True

wheel_vel	class_type	isaacsim.envs.mdp.actions.joint_actions:JointVelocityAction
	asset_name	robot
	debug_vis	False
	clip	None
	joint_names	FL_foot_joint
		FR_foot_joint
	scale	10.0
	offset	0.0
	preserve_order	True
	use_default_offset	False

[Rewards]

<i>reward</i>	<i>weight</i>	<i>param</i>
track_ang_vel_z_exp	1.5	command_name: 'base_velocity'
		std: 0.5
track_lin_vel_xy_exp	2.0	command_name: 'base_velocity'
		std: 0.25
joint_deviation_hip	-2.0	asset_cfg: {'name': 'robot', 'joint_names': ['.*_hip_joint'], 'joint_ids': slice(None, None, None), 'fixed_tendon_names': None, 'fixed_tendon_ids': slice(None, None, None), 'body_names': None, 'body_ids': slice(None, None, None), 'object_collection_names': None, 'object_collection_ids': slice(None, None, None), 'preserve_order': False}
joint_deviation_calf	-1.5	asset_cfg: {'name': 'robot', 'joint_names': ['.*_calf_joint'], 'joint_ids': slice(None, None, None), 'fixed_tendon_names': None, 'fixed_tendon_ids': slice(None, None, None), 'body_names': None, 'body_ids': slice(None, None, None), 'object_collection_names': None, 'object_collection_ids': slice(None, None, None), 'preserve_order': False}
dof_pos_limits_hip	-1.0	asset_cfg: {'name': 'robot', 'joint_names': '.*_hip_joint', 'joint_ids': slice(None, None, None), 'fixed_tendon_names': None, 'fixed_tendon_ids': slice(None, None, None), 'body_names': None, 'body_ids': slice(None, None, None), 'object_collection_names': None, 'object_collection_ids': slice(None, None, None), 'preserve_order': False}
dof_pos_limits_calf	-1.0	asset_cfg: {'name': 'robot', 'joint_names': '.*_calf_joint', 'joint_ids': slice(None, None, None), 'fixed_tendon_names': None, 'fixed_tendon_ids': slice(None, None, None), 'body_names': None, 'body_ids': slice(None, None, None), 'object_collection_names': None, 'object_collection_ids': slice(None, None, None), 'preserve_order': False}
dof_pos_limits_leg	-1.0	asset_cfg: {'name': 'robot', 'joint_names': '.*_thigh_joint', 'joint_ids': slice(None, None, None), 'fixed_tendon_names': None, 'fixed_tendon_ids': slice(None, None, None), 'body_names': None, 'body_ids': slice(None, None, None), 'object_collection_names': None, 'object_collection_ids': slice(None, None, None), 'preserve_order': False}
undesired_contacts	-0.5	sensor_cfg: {'name': 'contact_forces', 'joint_names': None, 'joint_ids': slice(None, None, None), 'fixed_tendon_names': None, 'fixed_tendon_ids': slice(None, None, None), 'body_names': ['base_link', '.*_thigh', '.*_calf', '.*_hip'], 'body_ids': slice(None, None, None), 'object_collection_names': None, 'object_collection_ids': slice(None, None, None), 'preserve_order': False}
		threshold: 1.0

joint_applied_torque_limits	-0.1	asset_cfg: {'name': 'robot', 'joint_names': '.*_joint', 'joint_ids': slice(None, None, None), 'fixed_tendon_names': None, 'fixed_tendon_ids': slice(None, None, None), 'body_names': None, 'body_ids': slice(None, None, None), 'object_collection_names': None, 'object_collection_ids': slice(None, None, None), 'preserve_order': False}
calf_align_l1	-0.2	asset_cfg: {'name': 'robot', 'joint_names': '.*_calf_joint', 'joint_ids': slice(None, None, None), 'fixed_tendon_names': None, 'fixed_tendon_ids': slice(None, None, None), 'body_names': None, 'body_ids': slice(None, None, None), 'object_collection_names': None, 'object_collection_ids': slice(None, None, None), 'preserve_order': False}
leg_align_l1	-0.1	asset_cfg: {'name': 'robot', 'joint_names': '.*_thigh_joint', 'joint_ids': slice(None, None, None), 'fixed_tendon_names': None, 'fixed_tendon_ids': slice(None, None, None), 'body_names': None, 'body_ids': slice(None, None, None), 'object_collection_names': None, 'object_collection_ids': slice(None, None, None), 'preserve_order': False}
base_height	2.0	temperature: 35.0
		default_height: 0.45
		asset_cfg: {'name': 'robot', 'joint_names': None, 'joint_ids': slice(None, None, None), 'fixed_tendon_names': None, 'fixed_tendon_ids': slice(None, None, None), 'body_names': None, 'body_ids': slice(None, None, None), 'object_collection_names': None, 'object_collection_ids': slice(None, None, None), 'preserve_order': False}
		sensor_cfg: None

[Events]

<i>event</i>	<i>property</i>	<i>value</i>
physics_material	func	isaacsim.envs.mdp.events.randomize_rigid_body_material
	params	asset_cfg: {'name': 'robot', 'joint_names': None, 'joint_ids': slice(None, None, None), 'fixed_tendon_names': None, 'fixed_tendon_ids': slice(None, None, None), 'body_names': ['*_hip', '*_thigh', '*_calf', '*_foot', '.*base.*'], 'body_ids': slice(None, None, None), 'object_collection_names': None, 'object_collection_ids': slice(None, None, None), 'preserve_order': False}
		static_friction_range: (0.5, 1.0)
		dynamic_friction_range: (0.5, 0.8)
		restitution_range: (0.0, 0.0)
		num_buckets: 64
	mode	startup
	interval_range_s	None
	is_global_time	False
	min_step_count_between_reset	0
randomize_com_positions	func	RL_D1h.tasks.manager_based.rl_d1h.mdp.events.randomize_com_positions
	params	asset_cfg: {'name': 'robot', 'joint_names': None, 'joint_ids': slice(None, None, None), 'fixed_tendon_names': None, 'fixed_tendon_ids': slice(None, None, None), 'body_names': 'base_link', 'body_ids': slice(None, None, None), 'object_collection_names': None, 'object_collection_ids': slice(None, None, None), 'preserve_order': False}
		com_distribution_params: (-0.05, -0.02)
		operation: add
	mode	startup
	interval_range_s	None
	is_global_time	False
	min_step_count_between_reset	0

add_base_mass	func	isaacsim.envs.mdp.events.randomize_rigid_body_mass
	params	asset_cfg: {'name': 'robot', 'joint_names': None, 'joint_ids': slice(None, None, None), 'fixed_tendon_names': None, 'fixed_tendon_ids': slice(None, None, None), 'body_names': ['base_link'], 'body_ids': slice(None, None, None), 'object_collection_names': None, 'object_collection_ids': slice(None, None, None), 'preserve_order': False}
		mass_distribution_params: (-0.75, 3.0)
		operation: add
	mode	startup
	interval_range_s	None
	is_global_time	False
	min_step_count_between_reset	0
reset_base	func	isaacsim.envs.mdp.events.reset_root_state_uniform
	params	pose_range: {'x': (-0.2, 0.2), 'y': (-0.2, 0.2), 'yaw': (-3.14, 3.14)}
		velocity_range: {'x': (-0.5, 0.5), 'y': (0.0, 0.0), 'z': (0.0, 0.0), 'roll': (-0.25, 0.25), 'pitch': (-0.25, 0.25), 'yaw': (-0.0, 0.0)}
	mode	reset
	interval_range_s	None
	is_global_time	False
	min_step_count_between_reset	0
reset_robot_joints	func	isaacsim.envs.mdp.events.reset_joints_by_offset
	params	position_range: (-0.2, 0.2)
		velocity_range: (0.0, 0.0)
	mode	reset
	interval_range_s	None
	is_global_time	False
	min_step_count_between_reset	0

push_robot	func	isaacsim.envs.mdp.events.push_by_setting_velocity
	params	velocity_range: {'x': (-0.1, 0.05), 'y': (-0.1, 0.1), 'z': (-0.1, 0.1)}
	mode	interval
	interval_range_s	10.0
		15.0
	is_global_time	False
	min_step_count_between_reset	0